

GREATER ROCHESTER INTL, NY, USA

WMO: 725290

Lat: 43.117N

Lon: 77.677W

Elev: 164

StdP: 99.37

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14768

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-14.1	-20.6	0.6	-15.7	-18.4	0.8	-13.0	13.7	0.5	12.2	-1.4	4.2	240	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	31.6	22.8	30.0	21.9	28.5	21.0	24.1	29.2	23.1	27.9	22.3	26.7	5.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	17.6	26.9	21.6	16.6	25.7	20.7	15.7	24.9	73.2	29.3	69.4	28.0	66.3	26.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.6	8.5	-19.8	33.6	3.5	1.5	-22.3	34.7	-24.3	35.6	-26.3	36.4	-28.8	37.5
			-20.1	25.6	3.4	1.1	-22.5	26.5	-24.5	27.1	-26.4	27.7	-28.8	28.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-3.7	-2.8	1.5	7.9	14.5	19.5	22.0	21.1	17.4	11.1	5.0	-0.2
	DBStd	10.26	6.53	5.83	5.93	5.10	4.71	3.73	3.06	2.97	4.02	4.62	4.97	5.12
	HDD10.0	1689	426	359	272	99	11	0	0	0	1	42	164	317
	HDD18.3	3558	682	591	522	314	138	30	5	8	64	228	400	574
	CDD10.0	1516	2	1	9	37	151	286	371	345	223	77	14	2
	CDD18.3	342	0	0	0	2	20	66	117	95	37	4	0	0
	CDH23.3	2774	0	0	4	32	200	554	982	691	278	32	0	0
	CDH26.7	785	0	0	0	6	50	162	306	185	73	3	0	0
Wind	WSAvg	4.0	4.8	4.7	4.6	4.5	3.9	3.5	3.3	3.1	3.3	3.8	4.2	4.4
Precipitation	PrecAvg	799	53	45	57	65	69	78	71	89	78	62	73	59
	PrecMax	1145	143	100	119	123	157	181	161	152	160	145	178	106
	PrecMin	534	15	14	25	30	9	6	16	21	7	6	11	20
	PrecStd	131	27	20	24	26	36	42	39	32	36	31	32	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	16.2	22.3	27.3	30.7	32.5	33.5	33.0	31.9	26.7	20.3	16.1
		MCWB	12.6	11.2	15.1	17.2	21.0	23.2	23.8	23.4	22.8	19.2	14.3	11.9
	2%	DB	11.9	11.6	17.1	23.1	28.0	30.4	31.7	30.5	28.7	23.6	17.8	12.0
		MCWB	9.8	7.7	11.6	14.7	19.4	21.8	23.0	22.5	21.5	17.6	13.1	8.9
	5%	DB	8.2	8.1	13.9	19.7	25.6	28.6	29.9	28.6	26.5	21.0	15.6	9.7
		MCWB	6.0	5.0	9.7	12.8	18.2	21.0	22.0	21.4	20.2	16.3	11.7	7.3
	10%	DB	5.1	5.0	10.5	16.7	23.0	26.6	28.2	27.1	24.4	18.5	12.8	7.0
		MCWB	3.1	2.7	6.6	11.1	16.6	19.9	21.0	20.7	19.1	14.4	9.4	4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	12.2	15.8	18.2	22.5	24.8	25.7	25.2	23.9	20.9	15.6	12.9
		MCDB	15.9	15.2	21.4	25.3	28.8	30.1	31.3	30.9	29.2	24.6	18.7	15.5
	2%	WB	9.8	8.2	12.5	15.8	20.7	23.2	24.3	23.7	22.6	18.6	14.0	9.7
		MCDB	11.7	11.3	16.2	21.2	26.0	28.3	29.4	28.3	27.2	22.3	17.0	11.4
	5%	WB	6.3	5.2	9.9	13.9	19.2	22.1	23.2	22.7	21.2	16.7	12.0	7.2
		MCDB	8.1	8.0	13.6	18.8	24.1	26.8	27.9	26.9	25.1	20.3	15.2	9.5
	10%	WB	3.2	2.7	6.8	11.6	17.7	21.0	22.2	21.7	20.0	15.2	9.9	4.8
		MCDB	4.9	4.8	10.1	16.1	22.0	25.1	26.6	25.6	23.5	18.2	12.6	6.8
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.1	8.7	10.3	10.9	10.4	10.4	10.1	10.3	9.1	7.9	6.6
		MCDBR	10.2	12.2	13.6	15.7	13.8	12.6	12.1	12.0	12.6	12.6	12.0	9.8
		MCWBR	8.9	9.6	9.2	9.2	7.2	6.1	5.5	5.6	6.3	7.6	8.4	8.1
	5% WB	MCDBR	9.8	11.5	12.9	14.0	12.3	10.9	10.2	10.1	10.9	11.3	10.6	9.2
		MCWBR	9.0	9.6	9.2	9.3	7.2	6.2	5.4	5.3	5.9	7.4	8.4	8.2
Clear-Sky Solar Irradiance	taub	0.306	0.316	0.335	0.371	0.395	0.436	0.433	0.421	0.374	0.357	0.313	0.294	
	taud	2.380	2.355	2.389	2.339	2.320	2.235	2.255	2.274	2.391	2.427	2.511	2.483	
	Ebn at Noon	825	880	906	895	879	840	838	835	853	822	818	814	
	Edh at Noon	79	96	105	120	125	137	133	126	104	89	70	66	
All-Sky Solar Radiation	RadAvg	1.29	2.09	3.34	4.48	5.47	5.81	5.89	5.24	4.18	2.52	1.56	1.03	
	RadStd	0.12	0.21	0.32	0.48	0.55	0.47	0.46	0.29	0.36	0.24	0.21	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+91	N/A
	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+71	N/A

Nomenclature: See separate page